



Department of Information and Communication Technology

Faculty of Technology

Rajarata University of Sri Lanka

ICT 1209 – Web Technologies | Laboratory 03 – Introduction to Web and HTML

Time: 120 Minutes

Intended Learning Outcomes (ILOs)

By the end of this practical, students should be able to

1. Apply CSS to a web page using inline, internal, and external methods.
2. Utilize different types of CSS selectors: element, group, universal, ID, and class to target specific HTML elements.
3. Implement basic styling properties (color, background-color, border, font-family, text-align) to enhance webpage presentation.
4. Utilize the concept of the CSS Box Model by applying margin and padding to control element spacing.

Pre-Practical Preparation

Reading/Resources:

1. Review the basics of CSS: W3Schools CSS Introduction
<https://www.w3schools.com/css/>
2. Review concepts related to the CSS Box Model (W3Schools)

Practical Objective

The goal of this practical is to introduce students to the fundamentals of CSS (Cascading Style Sheets). Students will learn the three main ways to apply styles, understand the role of different CSS selectors, and grasp the essential concept of the CSS Box Model to control element spacing and dimensions.

Submission

At the end of the practical, save all your work in a folder renamed with your student ID (IT2024_12345), compress it and upload it to the LMS.

Practical Activity

Task 1 - Applying CSS

Using Inline, Internal, and External CSS to apply CSS to a webpage.

- Objective: Apply styles to a webpage using all three primary CSS methods.
- Instructions:
 1. Create a new HTML file named *apply-css.html*.
 2. Add a heading and a paragraph with the following styles (as in Figure 1)
 - **Inline CSS** - Change the heading color to blue using the *style* attribute.
 - **Internal CSS** - Change the sub-heading text to red using a *<style>* block in the *<head>*.
 - **External CSS** - Link an external CSS file named *styles.css* to change the paragraph to green and background color of the page to light grey.

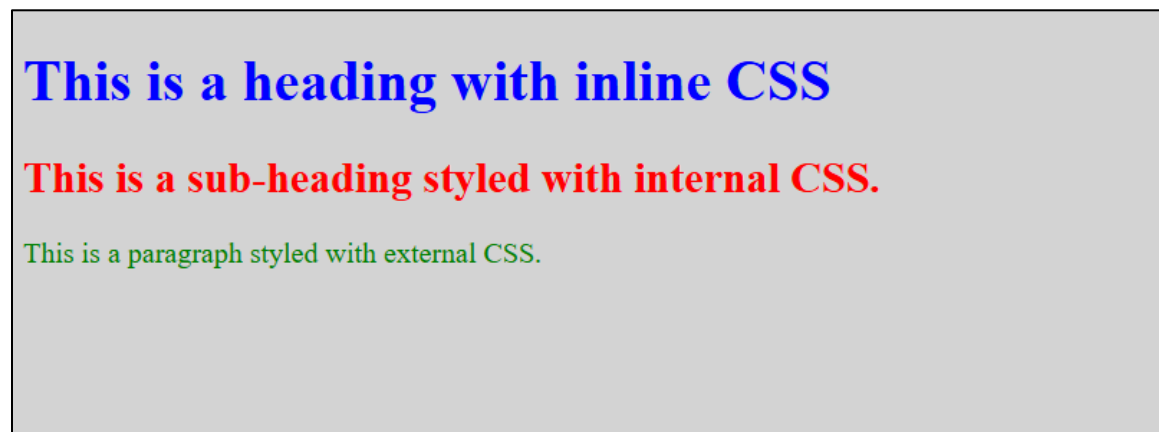


Fig. 1. Expected output of applying CSS to a webpage for Task 1

Sample Code - HTML (apply-css.html)

```
<!DOCTYPE html>
<html>
<head>
  <title>Insert Your Name</title>

  <!-- Internal CSS -->
  <style>
    h2 {
      color: red;
    }
  </style>

  <!-- External CSS -->
  <link rel="stylesheet" href="styles.css">
</head>
<body>

  <!-- Inline CSS -->
  <h1 style="color: blue;">This is a heading with inline CSS</h1>

  <h2>This is a sub-heading styled with internal CSS.</h2>
  <p>This is a paragraph styled with external CSS.</p>

</body>
</html>
```

CSS (styles.css)

```
p {
  color: green;
}
body {
  background-color: lightgrey;
}
```

Task 2 - Exploring CSS Selectors

Using different CSS Selectors target specific HTML elements

- Objective: Apply styles to elements using element, group, universal, class, and ID selectors.
- Instructions:
 1. Create a new HTML file named selectors_demo.html.
 2. Apply styles to different elements using
 - **Element Selector** - Change all <h1> tags to green.
 - **Group Selector** - Change all <p> and <h2> tags to italic.
 - **Universal Selector** - Set the font-family of the entire page to Arial.

- **Class Selector** - Apply a yellow background to a specific paragraph.
- **ID Selector** - Change the color of one specific heading to purple.

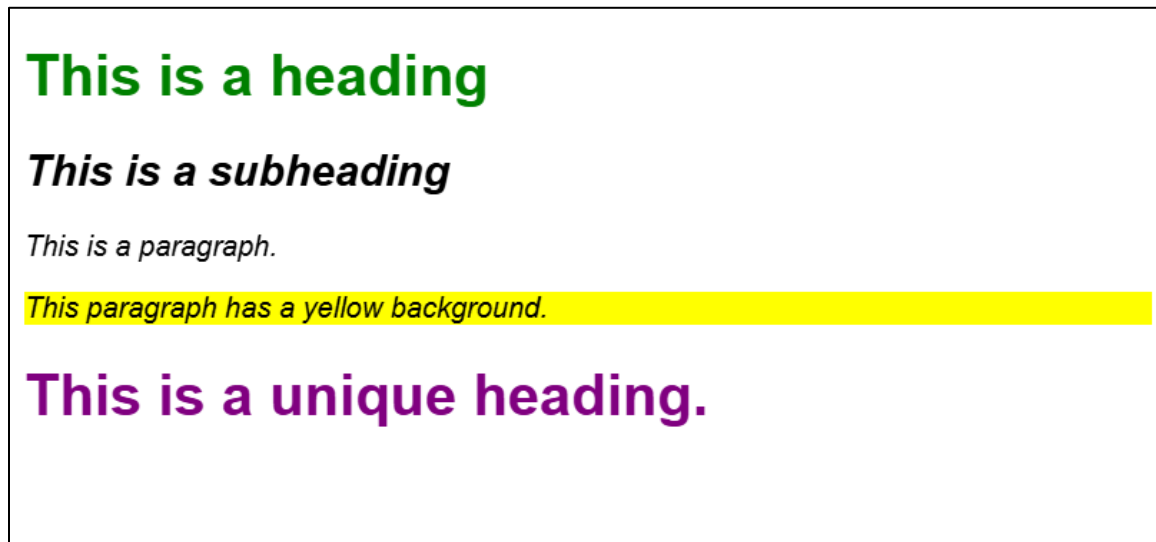


Fig. 2. Expected output of CSS selectors webpage for Task 2

Example Code - HTML (selectors_demo.html)

```
<!DOCTYPE html>
<html>
<head>
  <title>CSS Selectors</title>
  <link rel="stylesheet" href="selectors.css">
</head>
<body>
  <h1>This is a heading</h1>
  <h2 class="special-heading">This is a subheading</h2>
  <p>This is a paragraph.</p>
  <p class="highlight">This paragraph has a yellow background.</p>
  <h1 id="unique">This is a unique heading.</h1>
</body>
</html>
```

CSS (selectors.css)

```
/* Element Selector */
h1 {
  color: green;
}

/* Group Selector */
p, h2 {
  font-style: italic;
}

/* Universal Selector */
* {
```

```

    font-family: Arial;
}

/* Class Selector */
.highlight {
    background-color: yellow;
}

/* ID Selector */
#unique {
    color: purple;
}

```

Task 3 - Inline vs. Block Elements

Task 3 - Observing Default Element Behavior

- Objective - Differentiate between block-level and inline-level elements, and learn how to modify their default behavior using the display property.
- Instructions
 1. Create a new HTML file named display_demo.html.
 2. Use the following code snippet to add a few examples of common block elements (e.g., <div>, <h1>, <p>) and inline elements (e.g., , <a>,).

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Display Property Demo</title>
</head>
<body>
  <h2>Block Elements (Take up full width)</h2>
  <div>This is a DIV (Block Element)</div>
  <h1>This is an H1 (Block Element)</h1>
  <p class="block-to-inline">This is a Paragraph (Block Element) which will be changed to
  Inline in the task.</p>

  <hr>

  <h2>Inline Elements (Take up only necessary width)</h2>
  <p>
    This is a span element: <span class="inline-to-block">Span 1</span>
    and here is another: <span class="inline-to-block">Span 2</span>.
  </p>

```

```

<p>
  <a href="#">This is a Link (Inline Element)</a>
  and this text is <strong>Strong (Inline Element)</strong>.
</p>

</body>
</html>

```

3. Add a border to all elements so you can clearly see their boundaries.

Use universal selector for achieving this.

```

* {
  border: 1px dashed black;
}

```

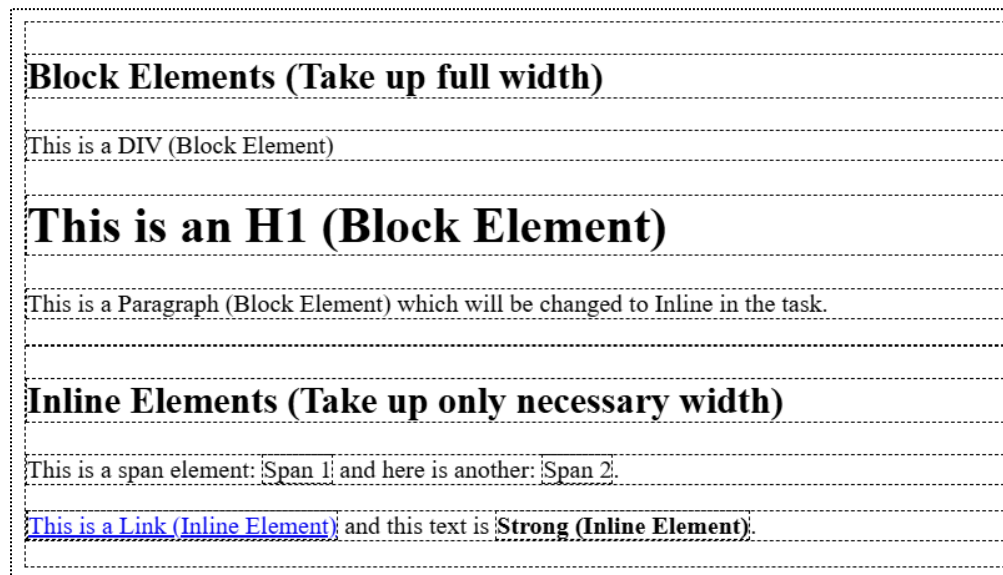


Fig. 3. Output for the task 3 after adding border for the elements

Observe the difference: Block elements take up the full available width and start on a new line, while inline elements only take up necessary width and flow on the same line.

4. Use CSS to change an inline element () to behave like a block element using display: block;.

```

.block-to-inline {
  display: inline;
  background-color: aqua;
}

```

5. Use CSS to make a block element (e.g., <h1>) behave like an inline element using display: inline;.

```
.inline-to-block {  
    display: block;  
    background-color: coral;  
}
```

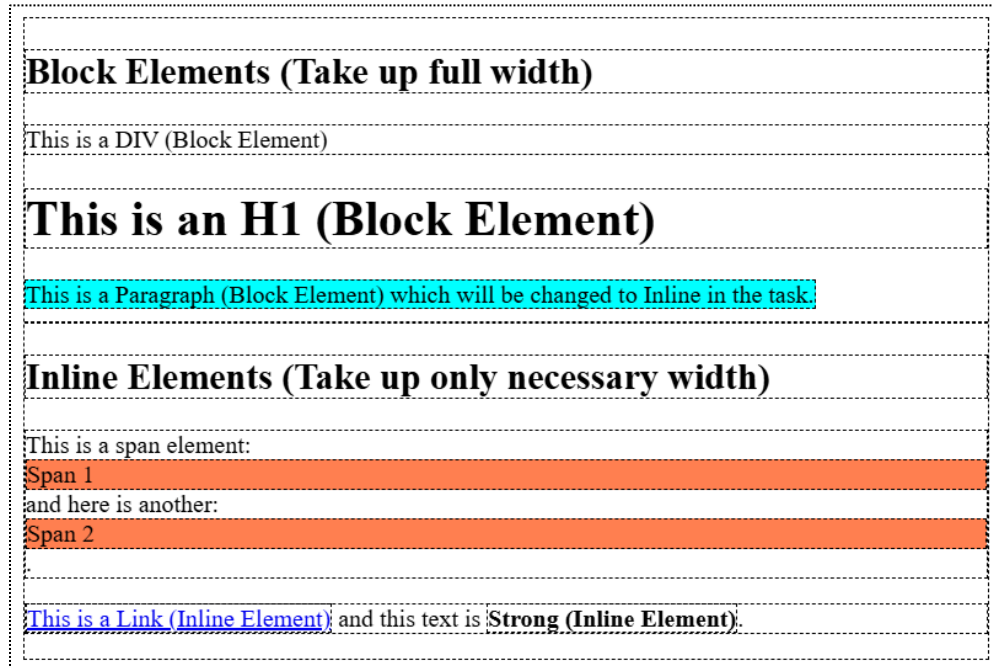


Fig. 4. Output for the task 3 after changing a block paragraph to inline element and inline span to block

Task 4 - Applying the CSS Box Model

Understanding Padding, Margin, and Border

- Objective - Demonstrate understanding of the CSS Box Model by correctly applying and manipulating padding, margin, and border properties to control element spacing and layout using the common <div> container.
- Instructions

Starter Code - HTML (box_model_demo.html)

```
<!DOCTYPE html>  
<html lang="en">  
<head>  
    <meta charset="UTF-8">  
    <title>Box Model Demo</title>  
    <link rel="stylesheet" href="box_model.css">  
</head>  
<body>
```

```
<div class="box box-1">
  <h2>Box 1: The Content Container</h2>
  <p>This div will be styled to show the complete Box Model.</p>
</div>

<div class="box box-2">
  <h2>Box 2: Adjacent Element</h2>
  <p>This box demonstrates how margins separate elements.</p>
</div>

</body>
</html>
```

1. Create an external CSS file named `box_model.css` and link it.
2. In `box_model.css`, apply the following styles to the `.box` class:
 - Set a fixed width of 300px and a height of 150px.
 - Apply a visible border (e.g., 10px solid darkblue).
3. Apply *display: inline-block* style to the `.box` class.

Note - Compared to *display: inline*, the major difference is that *inline-block* allows to set a *width* and *height* on the element.

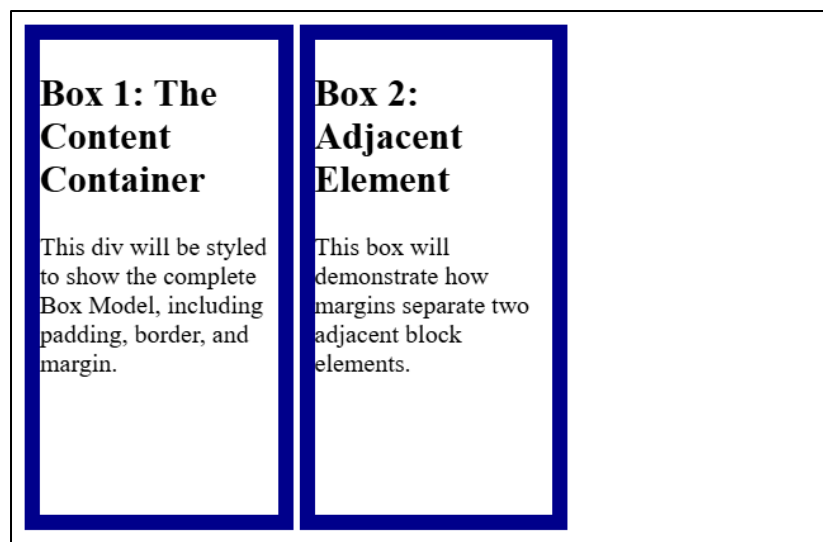


Fig. 5. Before adding margin or padding to inline-block elements in task 4

4. Set the background-color for `.box-1` to *lightgreen* to observe the separation created by Box 1's margin.

The .box-1 Class (Specific Styling)

- `background-color: lightgreen;` (Sets the background for the content and padding area.)
 - `margin: 10px 30px;` (Adds **external spacing** outside the border)
 - 10px margin on the top and bottom.
 - 30px margin on the left and right.
 - `padding: 10px 40px;` (Adds **internal spacing** between the content (text/images) and the border)
 - 10px padding on the top and bottom.
 - 40px padding on the left and right.
5. Set the *background-color* for .box-2 to *lightyellow* to observe the separation created by Box 2's **margin**.

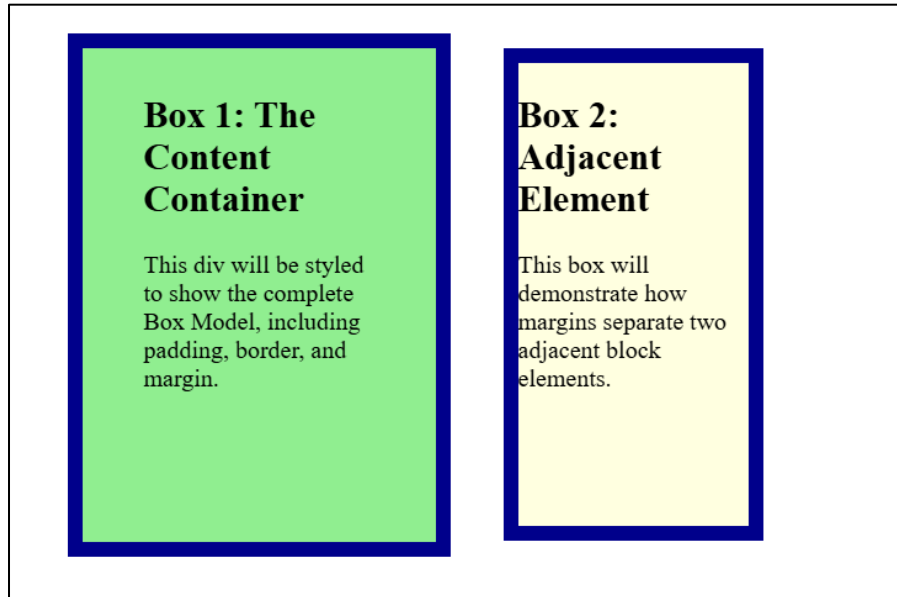


Fig. 6. Output for the task 4 showing use of the CSS box model applying and manipulating padding, margin, and border properties to control element spacing and layout

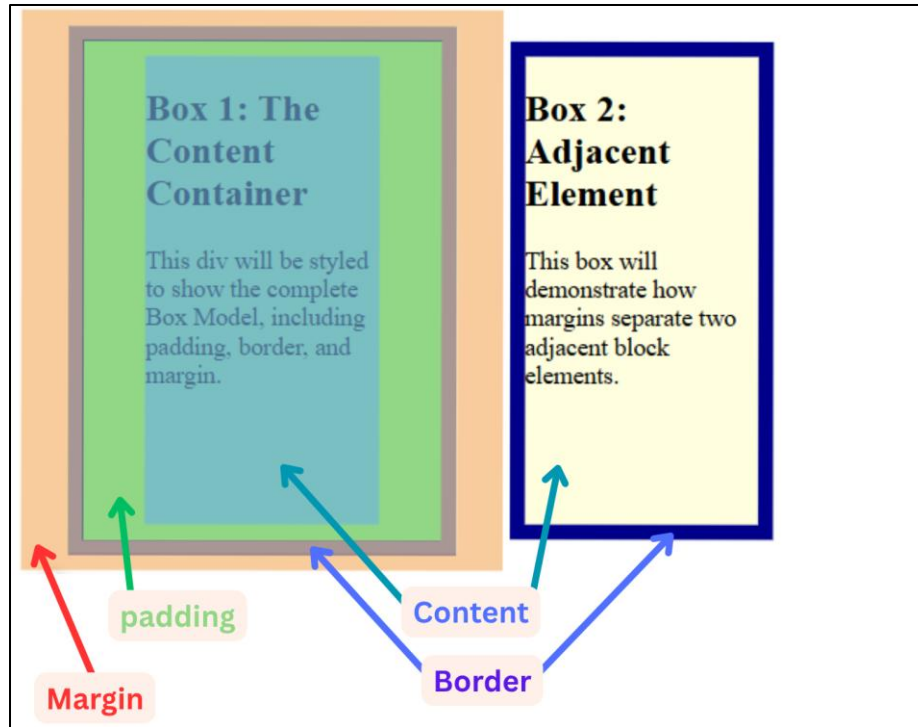


Fig. 7. Labelled image for the output for task 4 showing use of the CSS box model applying and manipulating padding, margin, and border properties to control element spacing and layout

You may see an (unwanted) small gap between adjacent inline-block elements. This gap is the browser rendering the whitespace (line break/space) in your HTML between the element tags.

Note: For true layout control without these issues in inline-block, use Flexbox or CSS Grid. These modern methods are the industry standard for component arrangement.

Challenge Task - Recreating a Page with External CSS

- Objective - Apply a range of CSS rules and Box Model properties to a complete HTML page using an external stylesheet, integrating concepts learned in previous tasks.
- Instructions
 1. Select a Starter Code: Choose either the "Mythical Theme" or the "Professional Theme" HTML code provided below to use as your profile.html file.
 2. External CSS - Create an external CSS file named profile_styles.css and link it to your chosen HTML file.
 3. Apply Styles - Use the CSS file to style the page according to the requirements:
 - Page Styling - Use the body selector to change the background color of the page.
 - Heading Styling - Use element and group selectors (h1, h2, h3) to style all headings with a different font (e.g., font-family: Arial;) and a specific color.

- Box Model on Paragraphs - Apply the Box Model by adding padding and margins to all <p> tags to separate them visually from surrounding elements.
- Advanced Styling (Optional) - Use a class or ID selector to change the background of a specific section or element.

If you don't have your last practical work with you, the same sample from earlier sheet is below.

Sample Starter Code for Mythical Theme

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>The Enchanted Grove</title>
  <meta charset="UTF-8">
</head>
<body>
  <div>
    <header>
      <h1>Welcome to the Enchanted Grove &#127794;</h1>
      <h3>Journey Through Myths and Magic</h3>
    </header>
    <section>
      <h2>Meet the Keeper of the Grove</h2>
      <p>Greetings, traveler! I am [Your Mythical Name], guardian of forgotten lore.</p>
      <ul>
        <li>Whispering to Trees &#127794;</li>
        <li>Reading Stars &#11088;</li>
        <li>Igniting the Flames &#128293;</li>
      </ul>
      
    </section>
    <section>
      <h2>Contact the Oracle</h2>
      <form>
        <label for="name">Your Name:</label>
        <input type="text" id="name" name="name" required><br><br>

        <label for="email">Your Email:</label>
        <input type="email" id="email" name="email" required><br><br>

        <label for="message">Your Message:</label><br>
        <textarea id="message" name="message" rows="4" required></textarea><br><br>

        <button type="submit">Summon Me</button>
      </form>
    </section>
  </div>

  <div>
    <footer>
      <p>May the stars guide you &copy; 2024 Enchanted Grove</p>
    </div>
  </div>
</body>
</html>
```

```

        <a href="https://www.example.com">Visit the Archives</a>
    </footer>
</div>
</body>
</html>

```

Sample Starter Code for the Professional Theme

```

<!DOCTYPE html>
<html lang="en">
<head>
    <title>John Doe - Web Developer</title>
    <meta charset="UTF-8">
</head>
<body>
    <div>
        <header>
            <h1>John Doe</h1>
            <h3>Aspiring Front-End Developer</h3>
        </header>

        <section>
            <h2>About Me</h2>
            <p>Hello! I'm John, a web development enthusiast. My goals</p>
            <ul>
                <li>Web Design</li>
                <li>Coding</li>
                <li>Creating Engaging User Experiences</li>
                <li>Exploring New Technologies &#128293;</li>
            </ul>
            
        </section>

        <section>
            <h2>Contact Me</h2>
            <form action="#" method="post">
                <label for="name">Your Name:</label>
                <input type="text" id="name" name="name" required><br><br>
                <label for="email">Your Email:</label>
                <input type="email" id="email" name="email" required><br><br>
                <label for="message">Your Message:</label><br>
                <textarea id="message" name="message" rows="4" required></textarea><br><br>
                <button type="submit">Send Message</button>
            </form>
        </section>
    </div>

    <div>
        <footer>
            <p>&copy; 2024 John Doe - All Rights Reserved</p>
            <a href="https://www.linkedin.com">Connect on LinkedIn</a>
        </footer>
    </div>
</body>
</html>

```